

EC824B

Double Selection & Incidental parameters problem

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1 Double Selection

2 Incidental parameters problem

Double Selection

We are interested in controlling for (high-dimensional) set of controls, Z ,

$$Y_i = D_i' \alpha_0 + g(Z_i) + U_i$$

For example [Belloni, Chernozhukov and Hansen (2014)]

- We are only interested in inference on α_0
- Want to control for Z as generically as possible, but maintain additive separability - this is one aspect of sparsity in this model

Double Selection

Model:

$$Y_i = D_i\alpha_0 + g(Z_i) + U_i \qquad \mathbb{E}[U_i|D_i, Z_i] = 0 \qquad (1)$$

$$D_i = g_D(Z_i) + V_i \qquad \mathbb{E}[V_i|Z_i] = 0 \qquad (2)$$

where the controls are high-dimensional in the sense that either:

1. many ($Z_i \in \mathbb{R}^K$ with $K \gg 0$), or,
2. consider set of functions of Z_i (e.g. B-splines, polynomials), or,
3. some combination of the two.

We would like to sufficiently control for $g(Z_i)$ or $g_D(Z_i)$ in either (1) OR (2)

Double Selection

- Call $X_i := P(Z_i)$, where $P(Z_i)$ are transformations of Z_i , e.g. B-splines, polynomials, etc.
- $g(Z_i) = X_i' \beta_g + r_{gi}$
- $g_D(Z_i) = X_i' \beta_D + r_{Di}$
- The estimation procedure works as follow

Step 1: Perform selection on regression of Y_i on X_i : get set of covariate indices I_1

Step 2: Perform selection on regression of D_i on X_i : get set of covariate indices I_2

Step 3: Post-selection regression of Y_i on D_i and X_{ij} for $j \in I_1 \cup I_2$

- The goal here is to control for regressors that effect Y_i or D_i - “robust”
- It can be shown that this has favourable properties to controlling only variables in the main regression of Y_i on D_i and X_i

Double Selection

Sparsity in the approximation errors:

- $g(Z_i) = X_i' \beta_g + r_{gi}$
- $g_D(Z_i) = X_i' \beta_D + r_{Di}$
- For $\|\beta_g\|_0 \leq s$, and $\|\beta_D\|_0 \leq s$
- $\Rightarrow \sum_{i=1}^N E[r_{gi}^2]/N \lesssim \sqrt{s/N}$, and $\sum_{i=1}^N E[r_{Di}^2]/N \lesssim \sqrt{s/N}$
- This last line is the sparsity requirement on the functions $g(\cdot)$ and $g_D(\cdot)$
- Sparser β_g, β_D imply smaller s , hence tighter bounds for r_{gi}^2 and r_{Di}^2
- Broadly, this implies simpler functions have faster convergence rates

Double Selection

- Selecting X_{ij} for $j \in I_1 \cup I_2$ (emphasising the union) gives a more “robust” set of controls
- For certain selection rules, the union provides strictly better inference properties than selecting only the intersection (you will see this in the problem set).
- In particular, we may have covariates that have small but non-zero β 's in the equation for Y_i , but are strongly correlated with D_i
- Inference here utilises classic inference procedures

1 Double Selection

2 Incidental parameters problem

Short T panels

As we have seen, fixed-effects in linear models can be dealt with in a simple fashion

$$Y_{it} = X_{it}\beta + \alpha_i + u_{it}$$

- Cross-sectional sampling: $(Y_{it}, X_{it}, \alpha_i, u_{it})$ are i.i.d. over i . Observe (Y_{it}, X_{it}) .
- Strict exogeneity (uncorrelatedness): $\mathbb{E}(X_{it}u_{is}) = 0$ for all $t, s \in \{1, \dots, T\}$
- “Fixed-effects”: No assumption on α_i , may be correlated with $X_{it} \Rightarrow$ not controlling for α_i leaves omitted variable bias
- **Question:** How to get an unbiased estimate for β ?

Answer: Eliminate α_i - derive model implications that do not depend on α_i , e.g.

$$Y_{it} - Y_{it-1} = (X_{it} - X_{it-1})\beta + u_{it} - u_{it-1}$$

Then use OLS

- How to generalise to non-linear models?

Short T panels

Classic semiparametric panel models

- Fully specified distribution: $u_{it} \sim i.i.d.F$ (e.g. probit or logit)
- Observe $Y_i = (Y_{i1}, \dots, Y_{iT})$ and $X_i = (X_{i1}, \dots, X_{iT})$
- Model:

$$f(Y_i|X_i, \alpha_i, \beta) = \prod_{t=1}^T f(Y_{it}|X_{it}, \alpha_i, \beta)$$

or with dynamics,

$$f(Y_i|X_i, Y_{i0}, \alpha_i, \beta) = \prod_{t=1}^T f(Y_{it}|X_{it}, Y_{it-1}, \alpha_i, \beta)$$

where α_i are unobserved fixed-effects, $\beta \in \mathbb{R}^{dim\beta}$ unknown common parameters

- Fully parametric model component: $f(Y_{it}|X_{it}, Y_{it-1}, \alpha_i, \beta)$
- Fully nonparametric model component: $f(\alpha_i|X_{it})$

Short T panels

- Fully parametric model component: $f(Y_{it}|X_{it}, Y_{it-1}, \alpha_i, \beta)$
- Fully nonparametric model component: $f(\alpha_i|X_{it})$

$$\Rightarrow \text{distr. of obs } f(Y_i|X_i) = \int f(Y_{it}|X_{it}, Y_{it-1}, \alpha_i, \beta_0) f(\alpha_i|X_{it})$$

- Goal: estimate β_0
- E.g. MLE with fixed-effects maximises log-likelihood over all parameters:

$$(\hat{\beta}, \hat{\alpha}_1, \dots, \hat{\alpha}_N) = \underset{\beta, \alpha_1, \dots, \alpha_N}{\operatorname{argmax}} \sum_{i=1}^N \log f(Y_{it}|X_{it}, \alpha_i, \beta)$$

- This estimator is inconsistent for fixed T and only $N \rightarrow \infty$
- [Neyman and Scott (1948)] *incidental parameter problem*

Short T panels

Examples of solutions to incidental parameter problem:

- Linear models: difference out FE, apply OLS/IV/GMM
- Sufficient statistic to eliminate FE
- Functional differencing: [Bonhomme (2012)]
- Maximum score: [Manski (1987)]
- Alternative asymptotic: $N, T \rightarrow \infty$, bias correction (what we look at the rest of this lecture)

Large T panels

“Nonparametric estimation”

- For every $i = 1, \dots, N$ observe,

$$(Y_{it}, X_{it}) \sim F_i(y, x), \quad t = 1, \dots, T$$

(data may be *i.i.d.*, or at least stationary and weakly correlated over time)

- As $T \rightarrow \infty$, we can *in principle*, estimate the conditional distribution $F_i(y_{it}|x_{it})$ separately for each $i = 1, \dots, N$. We can use this to estimate, e.g., the quantile function: $Q_i(\cdot|x) = F_i^{\leftarrow}(\cdot|x)$
- In the second step we can use $N \rightarrow \infty$ to learn the distribution of $Q_i(\tau|x_{it} = x)$ across i for every τ
- Thus, *in principle*, we can learn about the distribution of distributions, with unrestricted heterogeneity across i , as long as this is constant over t (i.e. we take the index i in α_i seriously).
- (In practice, N and T are finite, so we can only estimate a smaller number of individual specific parameters)

Large T panels

A parametric example

- Consider the linear regression with individual specific coefficients:

$$Y_{it} = X_{it}\beta_i + \alpha_i + \varepsilon_{it}, \quad \beta_i, \alpha_i \sim i.i.d.G(\beta, \alpha)$$

where G is the joint distribution of (β, α)

- For any $i \in \{1, \dots, N\}$, let $\hat{\beta}_i$ and $\hat{\alpha}_i$ be the OLS estimates from regressing Y_{it} on X_{it} over $t = 1, \dots, T$
- As $T \rightarrow \infty$ we have $\hat{\beta}_i \rightarrow \beta_i$ and $\hat{\alpha}_i \rightarrow \alpha_i$. Thus *in principle*, as $N \rightarrow \infty$ we can learn about the full distribution $G(\beta_i, \alpha_i)$

Large T panels

- We might be interested in the first and second moments,

$$m_k^{(1)} = \frac{1}{N} \sum_{i=1}^N \beta_{ik}, \quad m_k^{(2)} = \frac{1}{N} \sum_{i=1}^N \beta_{ik}^2$$

- With plug-in estimators,

$$\hat{m}_k^{(1)} = \frac{1}{N} \sum_{i=1}^N \hat{\beta}_{ik}, \quad \hat{m}_k^{(2)} = \frac{1}{N} \sum_{i=1}^N \hat{\beta}_{ik}^2$$

- Properties of $\hat{m}_k^{(1)}$ depend on if $\mathbb{E}\hat{\beta}_{ik} = \beta_{ik}$ (unbiasedness), which in panel data is only true if X_{it} is strictly exogenous
- For $\hat{m}_k^{(2)}$, very generally with $N, T \rightarrow \infty$ we have,

$$\text{bias}(\hat{m}_k^{(2)}) := \mathbb{E}\hat{m}_k^{(2)} - m_k^{(2)} = \frac{B}{T} + o\left(\frac{1}{T}\right), \quad \text{std}(\hat{m}_k^{(2)}) = \frac{C}{\sqrt{NT}} + o\left(\frac{1}{\sqrt{NT}}\right)$$

Large T panels

- For inference on $m_k^{(2)}$ using $\hat{m}_k^{(2)}$ we want,

$$\text{bias}(\hat{m}_k^{(2)}) \ll \text{std}(\hat{m}_k^{(2)})$$

but this is only true when $T \gg N$, which is not realistic in most examples
 $\Rightarrow$ need bias correction

- Claim: This discussion of asymptotic bias and standard deviation of $\hat{m}_k^{(2)}$ generalises to a very large class of fixed-effects estimators in panel data

Large T panels - Generalisations (aside from panel data)

Asymptotic with growing number of parameters:

- Samples size N ,
Unknown parameters $\theta = (\theta_1, \dots, \theta_p)'$
Corresponding estimator (e.g. MLE) $\hat{\theta} = (\hat{\theta}_1, \dots, \hat{\theta}_p)'$
- Conventional asymptotic: p fixed, $N \rightarrow \infty$. Then:

$$\sqrt{N}(\hat{\theta} - \theta^0) \xrightarrow{d} N(0, V_\theta)$$

- Alternative asymptotic: $N, p \rightarrow \infty$. Often:

$$\text{std}(\hat{\theta}_k) = O(N^{-1/2}), \quad \text{bias}(\hat{\theta}_k) = O(p/N).$$

- if $p \approx N^{-1/2}$ or $p/N \gg N^{-1/2}$

⇒ bias may be “first-order” relevant

Large T panels - Example: MLE with Fixed-Effects

- Specify $f(Y_i|X_i, \alpha_i, \beta)$, the conditional distribution of outcome variables $Y_i = (Y_{i1}, \dots, Y_{iT})$, e.g. for static binary choice model:

$$f(Y_i|X_i, \alpha_i, \beta) = \prod_{t=1}^T F(X'_{it}\beta + \alpha_i)^{Y_{it}} [1 - F(X'_{it}\beta + \alpha_i)]^{1-Y_{it}}$$
- Fixed-effect MLE maximises over all parameters:

$$(\hat{\beta}, \hat{\alpha}_1, \dots, \hat{\alpha}_N) = \underset{\beta, \alpha_1, \dots, \alpha_N}{\operatorname{argmax}} \sum_{i=1}^N \log f(Y_i|X_i, \alpha_i, \beta)$$

- Inconsistent for fixed T and $N \rightarrow \infty$ - Neyman and Scott (1948) *incidental parameter problem*
- $p = N + \dim \beta$ parameters and total sample size NT , thus p/NT does not converge to zero for fixed T .

Large T panels

- Consider again fixed-effect MLE maximises over all parameters:

$$(\hat{\beta}, \hat{\alpha}_1, \dots, \hat{\alpha}_N) = \operatorname{argmax}_{\beta, \alpha_1, \dots, \alpha_N} \sum_{i=1}^N \log f(Y_i | X_i, \alpha_i, \beta)$$

- Under alternative asymptotic, $N, T \rightarrow \infty$, consistency:

$$\operatorname{plim}_{N, T \rightarrow \infty} \hat{\beta} = \beta, \quad \operatorname{plim}_{N, T \rightarrow \infty} \hat{\alpha}_i = \alpha_i$$

- This follows from time-series arguments, i.e. $N \rightarrow \infty$ is not important for consistency
- ⇒ The alternative asymptotic, $N, T \rightarrow \infty$ transforms the *incidental parameter problem* into an *asymptotic bias problem*, and the problem has general solutions.

Large T panels

- Similar to the estimated moments, $\hat{m}_k^{(2)}$, from before, the panel fixed-effects estimators exhibit,

$$\text{std}(\hat{\beta}) \approx \frac{C}{\sqrt{NT}}, \quad \mathbb{E}(\hat{\beta} - \beta) \approx \frac{B}{T}$$

- Bias can be neglected only if $\frac{C}{\sqrt{NT}} \gg \frac{B}{T}$, i.e. $T \gg N$
 - This is usually not realistic, most applications have $N > T$, or, $N \approx T$
- ⇒ need an estimator with lower bias - *bias correction*

Large T panels

Analytic bias correction

- One can estimate the bias term B consistently then construct bias corrected estimator:

$$\hat{\beta}^{BC} = \hat{\beta} - \frac{\hat{B}}{T}$$

- For examples of $\hat{B}$, see [Hahn and Kuersteiner (2002)] for linear dynamic panel model, [Hahn and Newey (2004)] for nonlinear panel MLE
- Typically, for sufficiently large T,

$$\text{std}(\hat{\beta}^{BC}) \approx \text{std}(\hat{\beta}), \quad \mathbb{E}(\hat{\beta} - \beta) \approx \frac{B_2}{T^2}$$

⇒ remaining bias of order $1/T^2$ is negligible with respect to standard error if $\frac{C}{\sqrt{NT}} \gg \frac{B_2}{T^2}$, i.e.
 $T^3 \gg N$

Large T panels

Delete-One Jackknife Bias Correction

- Jackknife bias correction originally introduced by [Quenouille (1949, 1956)] in purely time-series or cross-sectional context
- Delete-one Jackknife for panel introduced by [Hahn and Newey (2004)]
- Elegant idea - only need to know that the leading bias for $\hat{\beta}$ of the form B/T , but do not need analytical estimate of B !
- Of course, this only works for static panels. Like the sample splitting earlier in the course, we cannot have dependence across indices otherwise ordering of the dataset matters (e.g. no AR(1))

Large T panels

Delete-One Jackknife Bias Correction

- Let $\hat{\beta}^{(T)} = \hat{\beta}$ be the original fixed-effects estimator (uses all $t = 1, \dots, T$
- Let $\hat{\beta}_{(-t)}^{(T-1)}$ be the same estimator, but without observations in t , i.e. using $T - 1$ time periods, N fewer data points. The average is,

$$\bar{\beta}^{(T-1)} = \frac{1}{T} \sum_{t=1}^T \hat{\beta}_{(-t)}^{(T-1)}$$

- The delete-one Jackknife bias corrected estimator reads

$$\hat{\beta}^{BC-2} = T\hat{\beta}^{(T)} - (T-1)\bar{\beta}^{(T-1)}$$

- Can show for sufficiently large T

$$\text{std}(\hat{\beta}^{BC-2}) \approx \text{std}(\hat{\beta}), \quad \mathbb{E} \left(\hat{\beta}^{BC-2} - \beta \right) \approx \frac{b_2}{T^2}$$

Large T panels

Split-panel jackknife bias correction

- Split-Panel Jackknife for panel introduced by [Dhaene and Jochmans (2015)] , also based on [Quenouille (1949, 1956)] idea
- Split-Panel Jackknife also works for dynamic models
- For simplicity assume T is even

Large T panels

Split-panel jackknife bias correction

- Again, let $\hat{\beta}^{(T)} = \hat{\beta}$ be the original fixed-effects estimator (uses all $t = 1, \dots, T$)
- Let $\hat{\beta}_1^{(T/2)}$ be the same estimator, but only using $t = 1, \dots, T/2$. Analogously, $\hat{\beta}_2^{(T/2)}$ for the second half, $t = T/2 + 1, \dots, T$. The split-panel average:

$$\bar{\beta}^{(T/2)} = \frac{1}{2} \left(\hat{\beta}_1^{(T/2)} + \hat{\beta}_2^{(T/2)} \right)$$

- The split-panel Jackknife bias corrected estimator reads

$$\hat{\beta}^{(BC-3)} = 2\hat{\beta}^{(T)} - \bar{\beta}^{(T/2)}$$

- Again, one can show that for sufficiently large T ,

$$\text{std}(\hat{\beta}^{BC-3}) \approx \text{std}(\hat{\beta}), \quad \mathbb{E} \left(\hat{\beta}^{BC-3} - \beta \right) \approx \frac{\tilde{b}_2}{T^2}$$

Large T panels

Higher order bias correction?

- Under regularity conditions (smoothness in the original model) can show that the bias of the fixed-effect panel estimator has a large T expansion of the form,

$$\mathbb{E}(\hat{\beta} - \beta) = \frac{B_1}{T} + \frac{B_2}{T^2} + \frac{B_3}{T^3} + \dots$$

- So far have shown how to eliminate the leading bias term, B_1/T . Higher order bias correction is possible, e.g. [Dhaene and Jochmans (2015)] extend the split-panel Jackknife, (assume T divisible by 6),

$$\hat{\beta}^{(BC-4)} = 3\hat{\beta}^{(T)} - 3\bar{\beta}^{(T/2)} + \bar{\beta}^{(T/3)}$$

This eliminates the first and second order bias, leaving $\frac{B_3}{T^3}$ order bias only

Large T panels

Higher order bias correction?

- *Warning:* For very large T , we still have

$$\text{std}(\hat{\beta}^{BC-4}) \approx \text{std}(\hat{\beta})$$

However, in practice with finite T , often,

$$\text{std}(\hat{\beta}^{BC-4}) \gg \text{std}(\hat{\beta})$$

called “variance inflation” due to bias correction

- Variance inflation might not be a big problem if the cross-sectional dimension, N , is sufficiently large so that $\text{std}(\hat{\beta}^{BC-4})$ is small. In practice, higher order bias correction is rarely used

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